BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Section "A" Multiple Choice Questions (MCQ's)

Maximum Marks: 12

Note: NOTE: (i) Attempt all the questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the question and its part according to the question paper. (iii) Each question carries 1 mark. (iv) Write the code of your question paper in bold letters in the beginning of the answer script.

Q1. Choose the correct answer for each question from the given option: (Model 1)

Q1. The branch of Chemistry which deals with hydrocarbons:

(A) Industrial chemistry
(B) Inorganic chemistry
(C) Organic chemistry
(D) Physical chemistry

Q2. If proton number is 19, electron configuration will be:

(A) (a) 2, 8, 9
(B) (b) 2, 8, 8, 1
(C) (c) 2, 8, 1
(D) (d) 2, 8, 3

Q3. 4th and 5th period in periodic table are known as:

(A) Short period
(B) Long period
(C) Normal period
(D) Very long period

Q4. Number of electrons lost by atoms of group IIIA equals:

(A) 1
(B) 2
(C) 3
(D) 4

Q5. Oxygen belongs to group VIA so number of electrons in its valence shell are:

(A) 3
(B) 4
(C) 5
(D) 6

Q6. The mobility of liquids is lesser than:

(A) Solids
(B) Gases
(C) Bose Einstein Condensate
(D) Plasma

Q7. Molarity of a solution which is prepared by dissolving 40g sodium chloride in 500cm^3 of solution is:

(A) 1.4M
(B) 1.5M
(C) 1.33M
(D) 1.35M

Q8. In Daniel cell _____ is used a cathode.

(A) Zn
(B) Cu
(C) Sn
(D) Pb

Q9. Which gas can turn lime water milky?

(A) O_2
(B) NO_2
(C) CO_2
(D) N_2

Q10. Which compound is known as lunar caustic?

(A) KNO_3
(B) $AgNO_3$
(C) NaOH
(D) $NaNO_3$

Q11. Which one is correct statement:

(A) (a) Oxidation occurs at cathode
(B) (b) Reduction occurs at anode
(C) (c) Reduction occurs at cathode
(D) (d) Ions lose electrons at cathode

Q12. What is the particle size in suspension?

(A) 10^3 nm
(B) 10^2 nm
(C) less than 10^3 nm
(D) greater than 10^3 nm

Section "B" Short Answer Questions

Maximum Marks: 30

> Note: NOTE: Answer any TEN questions from this section. Each question carries 03 marks. Answer should be clear and according to need. No extra detail is required.

Q2. Do you Know the historical background of chemistry.

Q3. What is atomic number of an oxygen atom which have 8 electrons and 8 protons?

Q4. State the periodic law.

Q5. Why an atom form chemical bond?

Q6. Why gases diffuses rapidly, Explain?

Q7. Define the terms: solution, aqueous solution, solute and solvent and give an example of each.

Q8. Define oxidation and reduction in terms of loss or gain of oxygen or hydrogen.

Q9. Classifying the type of elements into metals, non-metals and metalloids.

Q10. Explain electronegative character of non-metals.

Q11. Identify the Alloy. | Components | Alloy | | Cu-Zn | | | Cu-Al-Mg-Ni | | | Cu-Zn-Sn | |

Q12. Why gasoline does not dissolve in water?

Q13. How kinetic molecular theory differentiates states of matter?

Q14. Complete the chart: | Atomic Number | Number of protons | Number of electrons | Electronic configuration | Number of valence electrons | | 11 | 11 | 12 | 2,8,11 | 1 | | 12 | | | | | | 13 | | | | | | 14 | | | | | | 15 | | | | | | 16 | | | | |

Q15. Why ionization energy, electron affinity and electronegativity exhibit same trend in period and groups?

Q16. Write down the names of sub atomic particles their masses in a.m.u with their unit charges.

Section "C" Detailed Answer Questions

Maximum Marks: 18

> Note: NOTE: Attempt any THREE questions from this section. Each question carries 06 marks.

Q17. What do you mean chemical species, explain ion, molecular ion and free.

Q18. Discuss Rutherford' gold metal foil experiment in the light of structure of atom.

Q19. Discuss in detail the long form of periodic table.

Q20. Define ionic bond. Discuss the formation of sodium chloride (NaCl).

Q21. Discuss the property of evaporation in liquids? Which factors affects the evaporation process?